

Murray State University

How a Language Model Works

Language models - reasoning - agents - 20 minutes

You need: one browser and `index.html`. It runs offline. For projection use Settings -> Presentation mode. Start fresh: the untrained model is useful evidence.

One question for the session

What changes when a text predictor is given more structure, more checking, or tools? This page shows three bounded mechanisms: a tiny language model learning next-character prediction, arithmetic experiments with written work and checking, and a tool loop with a scripted action selector. Use Details for the deeper architecture and implementation notes.

"Keep asking what the program learned, what it is calculating now, and what part is a teaching fixture."

LLM - Language model - 6 minutes

Question

How can adjustable numbers turn recent text into a next-character prediction?

Activity

- 0:00 Open LLM.** Use Tokenize text and Inspect parameters on an untrained prompt. The numbers begin as random values.
- 1:30 Choose Train the model.** Read one untrained sample, then narrate the loss and samples as the first era completes. Use Try next-token prediction to compare likely next characters; change Temperature only after the audience sees the ranked choices.
- 4:30 Choose Trace attention.** Select a character and show which earlier positions receive weight for that prediction. It is a live intermediate value, not evidence that the model has a mind or intent.

Expected observation

Training changes the weights, loss declines, and the model's next-character distribution changes. Its samples can become more patterned while still being limited by one short, original story.

Teaching limit

This is a character-level transformer trained from scratch on a tiny corpus. It demonstrates a transformer learning mechanism, not general language ability or a claim about every frontier system.

Reasoning - Worked steps - 8 minutes

Question

How do written intermediate steps and checking change a small model's behavior on addition?

Activity

- 6:00 Open Reasoning and choose Train and compare.** Compare direct-answer training with worked column arithmetic. The models share core width, depth, and heads, but their contexts, seeds, and update counts differ to fit their different text formats.
- 8:30 Choose Compare exam answers.** Read a live attempt and an exam row. The useful comparison is the presence of written intermediate work, not a universal performance promise.
- 10:30 Optional slow extension: Try more attempts.** Run 1, 2, 4, and 8 attempts with the same frozen under-trained model. The arithmetic checker rejects malformed work and accepted chains vote. More attempts add inference-time computation, not training.

Expected observation

Written steps provide smaller targets for the model to predict. Sampling plus checking can improve the chosen answer even when weights stay fixed.

Teaching limit

The checker validates column arithmetic from the addends. The optional accepted-example round also retains chains matching the generated answer, then uses ordinary next-token updates on those examples. That is rejection-selected supervised fine-tuning, not a policy-gradient reinforcement-learning update.

Agents - Tool-using loop - 5 minutes

Question

What does a system need in addition to text prediction to take actions and use results?

Activity

- 14:00 Open Agents, then choose You choose the actions.** The calculator, search, reading, notebook, observations, and distractor consequences all run locally.
- 16:00 Choose Watch the scripted policy.** Point out the calculator's deliberate bad syntax and its genuine error message, then the later corrected call. Open the raw context to see what the action selector can read.
- 18:00 Choose Tiny - last 450 characters, Restart run, and Watch the scripted policy again.** The goal leaves the visible context and the automatic run stops.

Expected observation

Tool results become new text in the context. The small window changes what the selector can use, so the same task can fail when its goal is no longer visible.

Teaching limit

The action selector in this activity is a scripted policy, including the calculator mistake and retry; it is not a learned model. The loop, local tools, observations, and context truncation are the genuine mechanisms on display.

Optional slow experiments and close

Let Train the model and Train and compare continue while you discuss their live displays; each can take a few minutes. Try more attempts and Fine-tune on accepted examples are optional extensions. Do not promise a particular score or curve: the page uses seeded, bounded demonstrations.

"A language model predicts. Reasoning experiments add written work, checking, or extra attempts. An agent adds a loop and tools; in this activity, its action choice is deliberately scripted."

- Which claim came from a learned model, a fixed calculation, or a scripted policy?
- Why can extra attempts help without changing weights?
- What would need to change before this agent activity used a learned action selector?

Presenter controls

Guide opens from the header and on entry. Settings contains Open Presenter Notes, Presentation mode, and Reset. Reset returns the complete demonstration to its fresh state. Open printable guide opens the rendered copy of this same guide.

Bryant Harrison - Murray State University - How a Language Model Works - Offline, single-file demonstration - no account or runtime network